

Potential mechanisms and functional significance of aperiodic neural activity

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Human and animal electrophysiology often looks noisy and unstructured. While some clearer features exist in the form of bursty oscillations and event-related potentials, recent methodological developments have helped to uncover surprising structure in what otherwise looks like noise. This signal, referred to as aperiodic activity, has received substantial recent attention, examining its physiological generators and relationship to behaviour, cognition, development, ageing and disease. Here we examine the putative physiological basis of aperiodic activity, its relationship to other measures of neural activity and evidence for its functional and clinical relevance. Computational modelling and empirical evidence show that aperiodic activity has many neural origins, primarily postsynaptic transmembrane currents across populations of neurons. While several other signal statistics capture features similar to aperiodic measures, new methods have allowed researchers to more directly link aperiodic activity to underlying physiological processes. We discuss the implications of these findings for our understanding of cognition and disease, and highlight open questions for future research.

Electrophysiological recordings provide a lens for studying the biological origin of cognition. The neural field potentials recorded are dynamic, containing both rhythmic and arrhythmic activity. Aperiodic neural activity refers to the non-oscillatory, irregular fluctuations in neural field potentials. While neural oscillations have been extensively implicated in cognition and disease^{1,2}, the study of aperiodic activity has recently experienced a rapid surge in interest^{3,4}. Aperiodic activity was historically viewed as noise—a nuisance variable to be ignored or removed—but recent evidence suggests that it is functional, dynamic and implicated in a diversity of cognitive and physiological processes. Moreover, there is accumulating evidence that traditional analysis techniques conflate oscillations and aperiodic activity, motivating re-evaluation of the roles of these distinct processes^{5,6}.

Here we review recent research highlighting the functional importance of aperiodic activity, focusing on cognition, development and ageing, and clinical relevance. We also review the putative generative mechanisms of aperiodic activity, highlighting leading computational models and important caveats. Finally, we introduce

the communication through aperiodic dynamics framework, outlining a potential mechanism by which aperiodic activity might have a functional role in neuronal communication. (All code used for all analyses and plots are publicly available on GitHub at https://github.com/voytekresearch/review_functional_aperiodic.)

Defining aperiodic activity

Neural sources of aperiodic activity

Extracellular field potentials can be measured using invasive neural recording methods such as local field potential (LFP) recordings or, more indirectly, magneto- and electroencephalography (MEG and EEG). These techniques—hereafter collectively referred to as field potentials—measure aggregate activity across populations of neurons and can provide valuable information about network function. Although we will cover many sources that probably contribute to neural field potentials, and by extension EEG, the primary contributor is postsynaptic transmembrane currents in neurons originating from proximal and distal sources^{7–9}. These inward and outward transmembrane currents

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all roughly follow a double exponential form that can be described by the amount of time they take to rise and decay, known as the time constant¹⁰. For very fast, AMPA-mediated excitatory glutamatergic synapses, this time constant is very short (less than 5 ms¹¹). Inhibitory GABA_A currents are slower, with time constants in the 5–25-ms range¹². Excitatory NMDA, inhibitory GABA_B, and neuromodulatory currents can range from 50 to 250 ms or more^{11,13,14}. Because spiking in cortical neurons tends to be irregular, with highly variable interspike intervals¹⁵, these currents all tend to converge with irregular timing and integrate to give rise to the LFP (Fig. 1a,b). Exceptions occur with synchronized bursting states, pacemaker neurons and sustained oscillations^{16,17}, and other factors including synchronized dipoles contribute to EEG in particular¹⁸. (See Box 1 for a glossary of key terms in this Review.)

Because transmembrane currents are all of this roughly (double) exponential form, any power spectral (frequency domain) representation of them will always manifest such that power at the highest frequencies is exponentially less than power at the lowest frequencies (Fig. 1c,d). This Lorentzian form, as it is known, is a natural consequence of the mathematical operations performed when estimating the power spectrum. This phenomenon—where power exponentially drops off as a function of frequency—is broadly referred to as $1/f$. Many real-world phenomena give rise to $1/f$ -like spectra, ranging from speech, music and natural scenes to stock market values and X-ray emissions from black holes^{19–22}. While many interesting claims have been made regarding the apparent similarity between the $1/f$ -like nature of neural activity and $1/f$ -like processes in the real world—such as the brain existing in a metastable state at the edge of chaos²⁰—the $1/f$ -like nature of neural field potential data is a natural consequence of the exponential form of the transmembrane currents that give rise to them^{23,24}.

There is a direct mathematical equivalence between the time constant of an exponential process in the time domain, such as transmembrane currents, and the shape of its resulting power spectrum²⁵. Because transmembrane currents have different time constants, they will have different spectral forms that all follow the same basic pattern. The fastest currents, which have the shortest time constants, give rise to a spectral form that has relatively constant power across the lowest frequencies until a point at which power then begins to drop off. The point at which power begins to decrease is referred to as the knee frequency. At the limit, where the time constant approaches zero, the current becomes a delta function with equal power at all frequencies and a knee frequency at infinity. As this function stretches out and the currents become longer/slower, the knee occurs at a much lower frequency, and power above the knee drops off faster (Fig. 1c,d). The presence of a knee, or knees, suggests that the neural data are not scale free but rather have a preferred timescale or timescales. These are not oscillatory, rhythmic phenomena but rather show that there are non-oscillatory autocorrelations in the data.

Although postsynaptic transmembrane currents are the primary contributor to field potentials, other forms of neural activity also contribute. For instance, non-synaptic activity such as calcium spikes also generate field potentials in less predictable ways⁷. Additionally, spiking activity can ‘leak’ into the field potential, contributing to high-frequency power^{7,26,27}. It is important to consider these other potential signal sources when attempting to interpret what a given spectral form reflects, as there is no clear, direct one-to-one mapping between a given spectral form and a single physiological generator.

Measuring aperiodic activity

It is difficult to disentangle the theory of aperiodic activity from the methods used to measure it. This is because new methods for quantifying aperiodic activity and separating it from oscillations have helped to drive the increasing interest in examining it: you cannot study what you cannot measure. Because of this, it is important to understand the main ways through which aperiodic activity is measured, with the awareness that the measurement of aperiodic activity is different

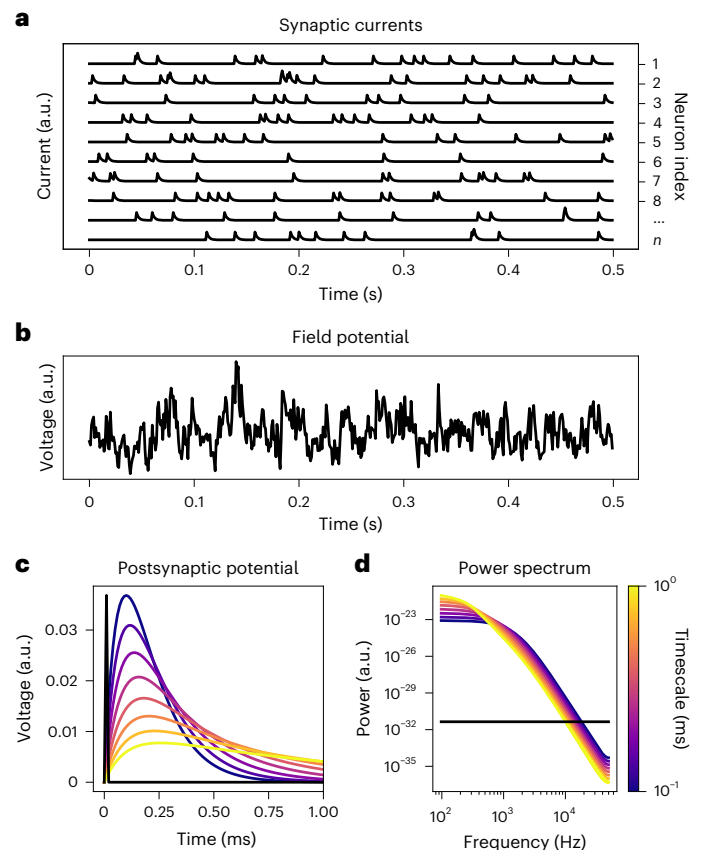


Fig. 1 | Basic model of aperiodic activity. **a, b**, Field potentials simulated as an integration (here, summation) of postsynaptic currents, resulting in an asynchronous signal visually resembling noise. Data simulated following refs. 23,24. **c**, Postsynaptic potentials of varying waveforms and timescales are simulated as double exponentials with varying widths but equal areas under the curve. Note that the delta function (black) has a width approaching zero. While not physiological, here it serves to guide mathematical intuition. **d**, The power spectrum of each simulated potential in **c** follows a similar form: power is relatively constant across frequencies until it decays exponentially (linearly in log–log space) after a knee frequency. The power of a delta function is constant with a knee at infinity.

from the underlying processes that generate it²⁸. Aperiodic activity is reflected in the power spectra of neural field potentials, which are the starting point for how most popular methods measure it. Neural power spectra can be parameterized by a small number of variables describing the oscillatory and aperiodic components of the spectrum⁵. In this approach, the power spectrum is modelled as the combination of an aperiodic component and oscillatory peaks:

$$P(f) = L(f) + \sum_{n=0}^N G_n(f) \quad (1)$$

where $P(f)$ is the power at frequency f , $L(f)$ is the aperiodic component and each $G_n(f)$ is an oscillatory peak. Each oscillatory component, $G_n(f)$, is modelled as a Gaussian function:

$$G_n(f) = a_n \exp \left[-\frac{(f - \mu_n)^2}{2\sigma_n^2} \right] \quad (2)$$

with amplitude a_n , centre frequency μ_n and standard deviation σ_n . The aperiodic component, $L(f)$, is modelled as a Lorentzian function:

$$L(f) = b - \log(k + f^\alpha) \quad (3)$$

BOX 1

Glossary

Local field potential (LFP): extracellular electrical signals recorded intracranially, reflecting subthreshold activity across populations of neurons.

Magnetoencephalography (MEG): a non-invasive recording technique used to record magnetic fields produced by the brain.

Electroencephalography (EEG): a non-invasive recording technique used to record electric fields produced by the brain.

Transmembrane currents: the movement of charged ions across the cell membrane, mediated by various ion channels and transporters.

Double exponential: the Laplace distribution, defined by two exponential functions.

Time constant: the time it takes a function to decrease by a factor of e .

Power spectral density: the squared magnitude of the Fourier transform.

1/f: used to describe a signal that has a power spectrum that is inversely proportional to frequency. The spectrum of neural signals typically follows a $1/f^\alpha$ distribution, where power exponentially decays as a function of frequency.

Knee frequency: the frequency at which power begins to decay as a function of frequency. In neural power spectra, frequencies below the knee typically have approximately equal power.

Poisson-distributed: having a discrete probability distribution defined by a constant mean rate and independent timing of events.

Event-related potentials (ERPs): stereotyped electrophysiological responses that occur at predictable times following sensory, cognitive or motor events.

with offset b , spectral knee k and aperiodic exponent X . The aperiodic component can alternatively be modelled as a linear function in log–log space, with parameter $k = 0$ (referred to as the fixed model).

The spectral exponent can also be measured using a number of other methods including irregular-resampling auto-spectral analysis (IRASA)²⁹. IRASA can be used to separate the oscillatory and fractal (aperiodic) components of the power spectrum so that the spectral exponent can be estimated from the aperiodic component. The algorithm involves resampling a time series by h and $1/h$, where h ranges from 1.1 to 1.9 in increments of 0.05; the power spectra of the resampled signals are then computed and averaged to estimate the aperiodic component, which can finally be fit with a power law. IRASA has the advantage of having few hyperparameters—namely, the resampling factors and those of the Fourier transform. One limitation is that IRASA assumes that the signal is fractal, while neural power spectra often exhibit features that violate this assumption, such as the knee driven by the autocorrelation, or timescale, of the neural signal (though newer implementations aim to address this³⁰).

Care must be taken when measuring aperiodic activity because there are many non-neural processes that also exhibit $1/f$ -like power law distributions. These sources must be considered and controlled for when performing analyses of aperiodic activity. These sources include non-physiological processes such as background noise, amplifier noise and electrode impedance. These sources can introduce white noise, which raises the noise floor and changes the shape of the power spectrum. The result is that estimates of both periodic and aperiodic activity parameters can be biased without explicitly controlling for them (for example, see ref. 23). This is especially pertinent when

comparing data collected from different populations or from different recording environments.

In addition to non-physiological processes, physiological noise sources such as muscle and ocular activity, respiration and the electrocardiogram (ECG) can also influence and bias estimates of aperiodic activity. While we refer to these physiological signals as ‘noise’, there are extensive, reciprocal connections between the central and peripheral nervous systems that will make disentangling these sources difficult. While the links between peripheral biosignal oscillations and neural activity are well known, such as with breathing³¹ and even the gastric basal rhythm³², the effects on aperiodic activity are unclear. Cardiac and respiration features also change as functions of cognitive and behavioural demands, which can make it even more difficult to discern between neural aperiodic and non-neural, physiological aperiodic activity changes, even in controlled tasks. What this means is that, in the strictest sense, the raw aperiodic parameters are uninterpretable without accounting for all of these noise sources—both physiological and non-physiological³³. To our knowledge, this has never been done. While certain recording modalities are less susceptible to some of these physiological noise sources—such as intracranial LFP or intracranial EEG (iEEG) recordings—they are not entirely immune to them.

It is therefore more common to instead report relative changes in aperiodic activity, such as in an event-related manner, or to compare two groups whose data were recorded in similar (or ideally the same) environments. Under these conditions, it is presumed that the noise sources are equivalent between task conditions or between groups. While this is probably true for non-physiological noise, this assumption is almost certainly incorrect for the physiological noise sources, as we noted. For example, it has been clearly demonstrated that subtle eye movements (microsaccades) can contaminate EEG in an event-related and task-dependent manner, giving rise to apparent neural gamma activity that is largely an artefact of the contamination by eye muscle activity³⁴. Similarly, it has been shown that even in the resting state, tonic differences in the ECG as a function of age can contaminate measures of apparent age-related differences in aperiodic neural activity³⁵. We thus recommend that studies of aperiodic neural activity control for these non-neural sources of noise as much as possible. When this is not possible (such as when reanalysing existing datasets that do not include these measures), more cautious interpretations are warranted.

Physiological evidence

The most commonly cited models of aperiodic activity assume that the LFP is predominantly a linear summation of excitatory and inhibitory synaptic currents induced by Poisson-distributed spiking activity, which has been shown to be a good approximation in detailed models of the LFP³⁶. This approach helps to link aperiodic activity to the underlying physiology^{23,24,37}. Using this approach, the spectral exponent has been specifically linked to excitation–inhibition balance²⁴.

While this prediction of this model has been confirmed empirically—wherein the spectral exponent of the LFP/EEG tracks excitation–inhibition balance^{24,38}—there is not a direct one-to-one mapping between the aperiodic exponent and excitation or inhibition³⁹. It is thus important to emphasize that the perspective that the aperiodic exponent captures excitation–inhibition dynamics holds true only under certain assumptions. Specifically, it assumes that ‘excitation’ is an AMPA excitatory process whose postsynaptic current has a very short, fast timescale; that ‘inhibition’ is a GABA inhibitory process with a longer, slower timescale; and that the currents are randomly distributed in time (not synchronized) and free of any oscillations. Any violations of these assumptions can break the purported relationship between excitation–inhibition balance and the spectral exponent.

For example, NMDA receptors are also excitatory, but their associated currents are longer and slower than GABA. This means that in an NMDA/GABA model of aperiodic activity, a steeper spectral exponent would index greater excitation, rather than the inverse relationship

seen in the traditional AMPA/GABA model. Furthermore, this model assumes the high-conductance state, in which the spiking activity of each neuron is uncorrelated, whereas neuronal networks often exhibit highly correlated activity across a range of cognitive and behavioural states^{40,41}. This correlated activity can contribute to increased spectral power, particularly at low frequencies¹⁸.

This can be demonstrated and visualized using the filter point process model of neural field potentials^{23,24,42} (Fig. 2). Here, neural field potentials are simulated under different degrees of spike synchrony (Fig. 2a). In the ‘desynchronized’ model, each spike train is modelled as a Poisson process, where the timing of each spike is independent of the next. In the ‘low-synchrony’ and ‘high-synchrony’ models, each spike train is a non-homogenous Poisson process, characterized by a time-varying rate. The rate of each spike train is coupled to a shared population rate that is modelled as an Ornstein–Uhlenbach process; in this model, the degree of spike synchrony can be controlled by modulating the coupling strength. Populations with greater spike synchrony exhibit steeper power spectra characterized by greater aperiodic exponents (Fig. 2b,c). This simulation was repeated across a range of coupling strengths, and linear regression was applied to test the relationship between the exponent and spike synchrony (quantified as the mean pairwise Pearson’s correlation coefficients) (Fig. 2c). This analysis revealed a positive correlation between the aperiodic exponent and spike synchrony ($r = 0.938$, $P = 2.03 \times 10^{-5}$).

In addition to population spike synchrony, there is clear evidence that oscillations can also alter aperiodic activity^{43,44}. While a stable oscillation manifests as a narrowband peak in the power spectrum and does not affect the $1/f$ -like component, if oscillatory power waxes and wanes, as is often the case in neural data, the rate of the power ramp also contributes a $1/f$ component (Fig. 2d). These results show that, while the simple model of aperiodic activity as the sum of asynchronous synaptic drive explains much of the $1/f$ power, the entirety of the $1/f$ -like signal is not derived solely from aperiodic processes.

Nevertheless, although there is no direct relationship between a given spectral exponent and excitation–inhibition balance, the predictions of this simple model have been at least partially confirmed in numerous studies^{24,38,45}. LFP recordings from rodent CA1 exhibit steeper power spectra (increased exponents) at greater recording depths, consistent with a greater density of inhibitory synapse density, and steeper spectra during the inhibitory phase of theta relative to the excitatory phase²⁴. Primates undergoing propofol anaesthesia (a GABA antagonist) exhibit increased aperiodic exponents relative to baseline, consistent with increased inhibition. In all cases, greater inhibition is associated with an increase in the aperiodic exponent.

The growing physiological evidence is such that we are getting a clearer picture about what can modulate aperiodic activity, under what conditions. For example, it is clear that aperiodic activity changes across layers of the hippocampus²⁴ and neocortex⁴⁶. There is also a growing body of causal, optogenetic work that is helping to clarify its source. One recent study tracked LFP and single-unit activity in the medial prefrontal cortex of mice over the first two postnatal weeks (P2–P12), a developmental period associated with increased inhibition³⁸. The authors found that the power spectrum grew steeper over this developmental period (increased exponent), an effect that was replicated in two human EEG datasets. The researchers then optogenetically manipulated inhibitory neuron activity at different developmental stages, finding that increased inhibition throughout development drives observed increases in the aperiodic exponent. Finally, in another study in mice, it was found that higher calcium activity predicted a flattening of aperiodic activity, while lower calcium activity predicted spectral steepening⁴⁵. The authors also showed that aperiodic activity was dynamically modulated by fluctuations in the ratio of pyramidal and interneuron activity.

While these results are exciting, there are many caveats and contrary findings. In the same calcium imaging study⁴⁵, the authors also

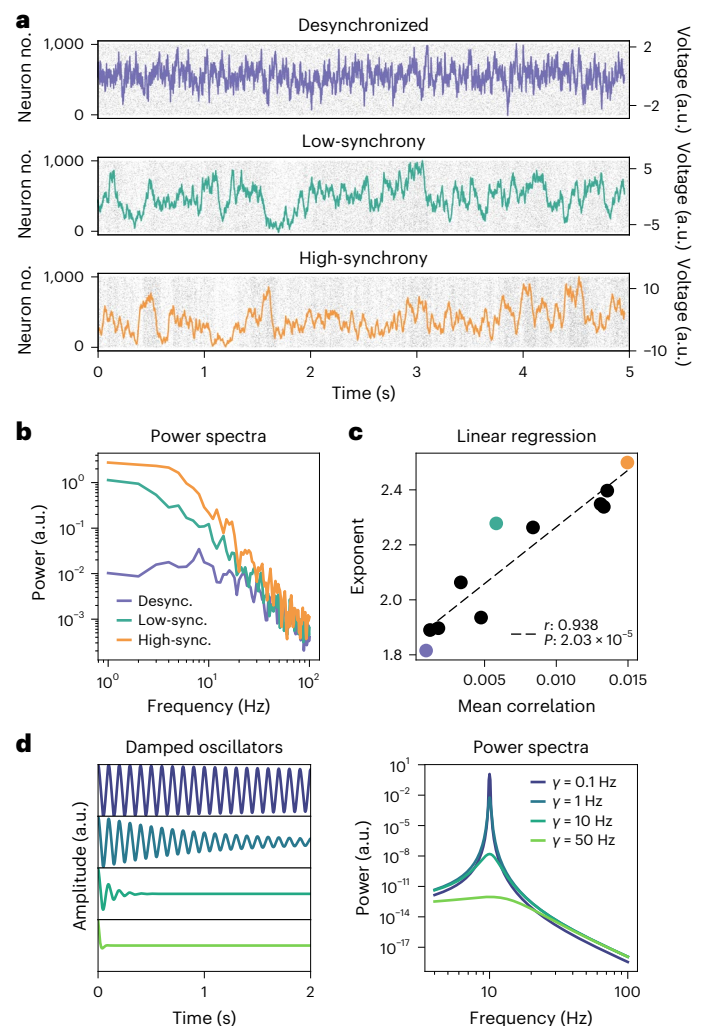


Fig. 2 | Additional neural processes that may influence aperiodic measures.

a–c, Neural time series simulated by filtered point process with varying spike synchrony. In the desynchronized model, each spike train is a Poisson process (**a**). In the low-synchrony and high-synchrony models, spike trains are non-homogenous Poisson processes coupled to a time-varying shared population rate (**a**). Synchronous populations exhibit greater total power (area under the curve) (**b**). The aperiodic exponent of the simulated field potential increases with its mean spike correlation (**c**). Linear regression results (dashed line), Pearson’s coefficient (r) and the P value are reported. See refs. 18,93,94 for demonstrations in more detailed models. **d**, Relationship between the aperiodic exponent and the decay rate (γ) of a damped oscillator, defined as $\cos(2\pi f_c t) \exp(\gamma t)$. Data simulated following ref. 43.

found that excessive pyramidal cell activity, indicative of very high excitatory activity, was accompanied by a steepening of the spectral exponent, which is exactly the opposite of what the simple excitation/inhibition model of aperiodic activity predicts. However, as seen in Fig. 2b, it is not just the amount of activity but also the correlations between units that can drive the spectral exponent. These results demonstrate the complexity of the relationship between the aperiodic exponent and excitation–inhibition balance.

Using a connectivity-based measure of excitation–inhibition balance, Diehl and Redish show that excitation–inhibition balance is also strongly related to broadband spectral power³⁹. Contrary to the simple model, the authors found a weak positive correlation between the aperiodic exponent and excitation–inhibition balance. In another recent study, Salvatore et al. performed several pharmacological manipulations of excitation–inhibition balance, including intraperitoneal drug injections and DREADDs (designer receptors exclusively activated

by designer drugs)⁴⁷. While some results align with the excitation–inhibition balance model, others do not. Specifically, the effects of picrotoxin, ketamine and DREADDs did not align with the predictions of the model. The authors found no effect of ketamine on aperiodic spectral components, a result that may be explained by the multifaceted mechanism of action of this drug. Picrotoxin (a GABA antagonist) unexpectedly steepened the exponent but also unexpectedly caused sedation, highlighting the complexity of drug effects. Furthermore, suppressing parvalbumin interneurons with DREADDs (disinhibition) unexpectedly resulted in an increased (steeper) aperiodic exponent.

These results show that there are many factors to account for when attempting to interpret aperiodic activity. As we have noted, the LFP primarily reflects inputs to a given region, so measuring local spiking outputs accounts for only a small portion of the LFP. More sophisticated biophysical models that account for many simultaneous processes, including the strength and identity of postsynaptic currents, recurrent connectivity, spike synchrony and so on, are yielding promising insights, providing a more complete map between a power spectral form and its physiological generators^{42,48}. However, in addition to better models, additional experimental studies are also needed. Simplified, *ex vivo* recordings in cortical slices could provide insight into microscale mechanisms of aperiodic activity changes in LFP activity. In *vivo*, high-density invasive recordings now allow researchers to simultaneously record spiking and LFP activity across brain regions with known connectivity patterns. Thus, it might be possible to record spikes from visual thalamic output regions along with visual cortical LFP to test the prediction that dynamic shifts in excitatory drive and synchrony shift the aperiodic exponent.

Integrating previous findings with aperiodic theory

Given the fact that aperiodic activity is dynamic, it is important to revisit and recontextualize previous work under the umbrella of aperiodic theory. For example, it is now well known that aperiodic dynamics can be mistaken for narrowband oscillatory changes^{5,6,49,50}. While there is ample evidence for the functional role of oscillations, especially with regard to rodent hippocampus theta⁵¹ and primate visual cortical alpha rhythms⁵², it is becoming clear that not every brain region oscillates at all times. The emerging perspective is such that even ‘classic’ oscillation effects, such as the role of human frontal midline theta⁵³ or oscillatory functional connectivity in EEG⁵⁴, are at least partly, if not largely, explainable by aperiodic activity as opposed to the presumed oscillations. Even when oscillations are present, it is likely that oscillations and aperiodic activity are not independent but interact with one another⁵⁵.

While not explicitly referring to them as aperiodic, a large body of literature has examined task-related changes in neural variability in time^{56,57}. In these studies, task events (such as the onset of a stimulus) are associated with a decrease in variability. We suggest that these dynamic changes could reflect shifts in aperiodic activity. Specifically, task-evoked decreases in neural variability are consistent with task-evoked decreases in the spectral exponent (that is, a flattening of the power spectrum). This is because signal variance is directly proportional to the integral of the power spectrum, and electrophysiological signals tend to pivot in the 20–50-Hz range^{24,50,58}. In simpler terms, when aperiodic activity flattens, power in the lowest frequencies, which have exponentially higher power and thus greater time domain variance, drops. So although power in higher frequencies also increases, the time domain peak-to-peak voltage fluctuations are relatively much lower, resulting in less neural variability.

This phenomenon can be clearly demonstrated in simulation (Fig. 3). Here, neural field potentials were simulated for 100 trials. On each trial, the signal exhibited a sudden shift in the aperiodic exponent at time 0 s (Fig. 3a). While the average exponent shifted from 1.5 to 1.0, the average trial-to-trial variability shifted from 1.0 to 0.6 (Fig. 3b,c). This simulation was repeated with a range of aperiodic exponents, showing a perfectly monotonic inverse relationship between the aperiodic exponent and trial-to-trial variability (Fig. 3d).

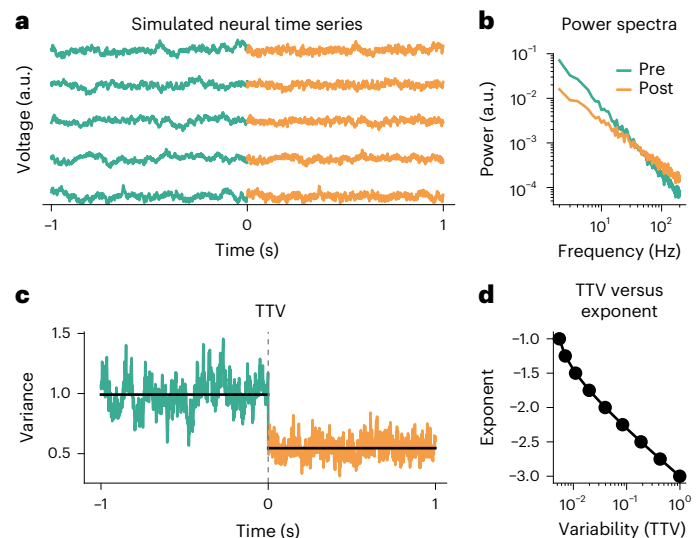


Fig. 3 | Relationship between trial-to-trial variability and the aperiodic exponent. Neural time series were simulated for 100 trials. On each trial, the aperiodic exponent suddenly decreased following stimulus onset (that is, time 0 s). **a**, Simulated neural time series for several example trials. **b**, Average power spectral density for the pre-stimulus (green) and post-stimulus (orange) time windows. **c**, Trial-to-trial variability (TTV) across time. **d**, TTV versus the aperiodic exponent across several simulations with varying aperiodic exponents.

Furthermore, we believe that several other measures such as multiscale entropy⁵⁹ and Lempel–Ziv complexity⁶⁰ capture highly overlapping, if not the same, dynamics as aperiodic activity. Recent studies for systematically comparing the relationships between many of these measures corroborate this. In one study, we applied several methods—including autocorrelation measures, spectral measures, detrended fluctuation analysis and several measures of fractal dimension, complexity and entropy—to simulated and empirical neural data, and found that these methods largely capture the same features⁶¹. In another study, we took a larger, data-driven approach, examining over 6,800 time-series features in a large MEG dataset, and found that time-series features mainly captured power spectral/timescale parameters⁶². Moreover, these timescale parameters followed the gradients of cortical micro-architecture, including gene expression and the density of neurotransmitter receptors and transporters. Continuing to connect these signal statistics to underlying physiology and behaviour will inform their utility in the study of cognition and disease.

An alternative interpretation of aperiodic activity is that of self-organized criticality²⁰. This general principle has been applied to physical systems other than the brain that also exhibit $1/f$ -like noise. Under this framework, a dynamic system with extended degrees of freedom will naturally evolve to reach a minimally stable attractor state characterized by power law scaling. In the context of electrophysiology, it is believed that the brain dynamically forms a network of interconnected subnetworks, so any perturbation to the system propagates through the network at all spatial scales, leading to temporal fluctuations with $1/f$ spectral properties. However, this interpretation is inconsistent with neural power spectra, which often have a characteristic timescale (the knee, an autocorrelation) that is physiologically relevant²⁵. Another challenge to the self-organized criticality interpretation is that criticality represents an unstable point that is sensitive to perturbations, while the brain exhibits robust aperiodic dynamics despite constant perturbation such as sensory inputs. Due to these inconsistencies, we do not believe that the $1/f$ -like nature of neural field potentials is in and of itself indicative of self-organized criticality.

It is worth mentioning that other non-oscillatory neural dynamics have been described that do not necessarily fall under the umbrella of

aperiodic activity. Aperiodic activity is distinct from transients such as event-related potentials (ERPs). ERPs can, however, contribute to the spectral form of field potentials, potentially obscuring measures of aperiodic activity. One must exercise caution when performing event-related analyses of aperiodic activity and check for any potential spectral contamination from ERPs, which typically contribute to low-frequency spectral power⁶³.

Functions of aperiodic activity

Aperiodic activity and cognition

There is rapidly accumulating evidence that aperiodic activity is functionally relevant across a variety of cognitive tasks. In one early example, we built off work on the steady-state evoked potentials, which are a specific type of ERP wherein rhythmic stimuli (typically a flickering light or rhythmic sound) at a particular frequency elicits a neural oscillation—or at least a rhythmic ERP—at that frequency. We showed that such temporal alignment of brain activity to sensory stimuli extends beyond these periodic stimulations to 1/f-like visual and auditory stimuli⁶⁴. We found that the spectral exponent of EEG signals tracked that of visual and auditory amplitude modulated noise in the respective sensory cortices. This effect was modulated by attention and shown to be predictive of behavioural performance.

As the popularity of aperiodic analyses has grown, there has been a rush to examine under which conditions and states aperiodic activity differs. Many other studies have likewise shown that aperiodic activity is modulated by attention and relevant to behaviour, yet there is not a clear mapping between aperiodic measurements and attention. The exact relationship seems to be dependent on task, participant age⁶⁵ and even participant education⁶⁶. However, one consistent finding in studies measuring aperiodic activity on a single-trial basis is that greater aperiodic exponents (steeper spectra) during the pre-stimulus period are associated with greater error rates^{67,68}.

Our first study of aperiodic dynamics showed that there were prominent differences in the aperiodic exponent between younger and older adults, with older adults showing a flatter spectrum. At the time, we interpreted this from a ‘neural noise’ perspective^{69,70}, because a flatter spectrum is more like a white-noise power spectrum, which arises from time-series data that are temporally uncorrelated. More recent work shows that working memory accuracy is also associated with the aperiodic exponent and timescale of iEEG. iEEG was recorded from humans performing a visuospatial working memory task, and the timescale of aperiodic activity in all analysed regions (including the parietal lobe, prefrontal cortex, orbitofrontal cortex and medial temporal lobe) was found to be around 20% longer during delay periods than at baseline. Furthermore, the event-related expansion of timescales in the prefrontal cortex was correlated with better working memory performance²⁵. These findings suggest that aperiodic activity has a functional role in working memory.

There is also increasing interest in the temporal dynamics of aperiodic activity during cognitive tasks. This work shows task-related modulation of aperiodic activity independent of oscillations and ERPs^{50,58,71,72}. The aperiodic exponent in iEEG decreases during sensory and motor responses in sensorimotor tasks⁵⁰ and during memory encoding⁷² and subsequent memory tasks⁵⁸. Furthermore, aperiodic activity was found to encode spatial location during working memory tasks with dynamics that are distinct from oscillations⁷². These findings demonstrate that aperiodic activity is a dynamic process with functional implications on the timescale of behaviour and cognition.

Communication through aperiodic dynamics

The mechanistic interpretation of aperiodic dynamics remains an open question. While we often use neural signals such as ERPs and oscillations as biomarkers for cognition, their mechanistic role in information processing is likewise inconclusive. Considering aperiodic activity as a functionally relevant signal will help us to disentangle its dynamics

from other features with which it is often conflated. But importantly, once we are able to more accurately measure a neural signal, we can then conduct more accurate experiments to understand its physiology.

While many processes drive aperiodic activity—such as dampened oscillators⁴³ and neuronal population firing synchrony¹⁸, as we showed above—aperiodic dynamics can still potentially be leveraged to build and test hypotheses about the physiology of cognition. For example, under scenarios where no oscillations are present and neuronal firing is presumed to be more asynchronous, we can be more certain that aperiodic activity is driven by the relative contributions of postsynaptic currents. While we might have to control for non-neural sources, such as event-locked autonomic changes⁷³, we may still be able to make informed inferences about the postsynaptic drive.

For example, given that most real-world auditory visual stimuli are asynchronous with different temporal and spatial autocorrelations, it is unsurprising that they drive cortex in an aperiodic but autocorrelated manner. This is similar to how visual stimuli with a characteristic frequency, or ‘spatial oscillations’ such as gratings and Gabor patches, give rise to visual cortical oscillations⁷⁴. We argue that strong, aperiodic autocorrelations in the neural signals increase the chance that two regions will be correlated. While perhaps unintuitive, it is a fact that any two random signals that each have an autocorrelation structure to them, even if the signals are independent from one another, will become increasingly correlated with one another as their autocorrelations increase⁷⁵. While often interpreted as spurious, this might instead be a statistical quirk that evolution has taken advantage of.

Along these lines, Wolff and colleagues recently proposed that the hierarchy of intrinsic neuronal timescales that is evident across modalities^{25,76,77} is central to how sensory inputs are segregated and temporally integrated⁷⁸. Here we expand on this idea under the assumption that aperiodic activity reflects population postsynaptic timescales, but we update the ‘intrinsic’ assumption to one where neuronal timescales are functionally dynamic and determined by the aggregate postsynaptic drive. Building off the foundational communication-through-coherence hypothesis regarding the functional role of neural oscillations⁷⁹, we argue that the temporal autocorrelation of a neural population can be temporarily expanded or contracted depending on the nature of the inputs. When the postsynaptic drive is dominated by longer-timescale inputs, the correlation structure between distant brain regions increases, facilitating interregional network communication.

Under this complementary communication-through-aperiodic-dynamics hypothesis, when regions dynamically shift from the short timescales driven by faster excitatory glutamatergic inputs to slower timescales, such as excitatory NMDA or neuromodulatory influence, the increase in the population autocorrelation increases correlations with other regions with similarly high autocorrelations. This phenomenon might potentially help to coordinate information transfer between brain regions by increasing their shared correlation structure. Population activity dominated by slower timescales has the appearance of being more like ‘slow waves’ even in the absence of any oscillations^{80,81}, having a similar appearance to up states and down states.

Aperiodic activity in development and ageing

Aperiodic activity has also been widely implicated in development and ageing. Strong evidence suggests that changes in aperiodic activity may underlie age-related cognitive decline. EEG flattening with age in adulthood has been consistently reported across several studies^{69,82–87}. Moreover, these changes have been linked to cognition; for instance, age-related differences in aperiodic activity were linked to response variability in a spatial attention task⁸⁶ and accuracy in a visual working memory task⁸⁷. One recent study found no relationship between aperiodic activity and age; however, the authors did find that aperiodic activity is associated with cognitive performance⁸⁸.

Although age-related changes in aperiodic activity have been widely replicated, Schmidt et al. argue that such changes may at least

partially reflect changes in cardiac activity³⁵. Both M/EEG and ECG spectra flatten with age, but ECG is rarely corrected for in M/EEG studies. The authors found that age-related differences in aperiodic M/EEG are better explained by cardiac, as opposed to neural, sources. This research is complicated by differences in results between recording sites and frequency ranges, however. The EEG exponent (particularly in the low frequency range) was related to heart-rate variability metrics. Importantly, ECG activity remained in M/EEG even after removal with independent component analysis. Using independent component analysis, the authors show that changes in the spectral exponent are more common in ECG components of MEG than in MEG or MEG with these components removed. This finding motivates further research on the relationship between brain and cardiac activity.

Research on the relationship between aperiodic activity and development is somewhat inconsistent and difficult to interpret due to differences in age ranges between studies and how age is measured (that is, chronological age versus postmenstrual age). We reported the progressive flattening of the EEG exponent in early development (1–7 months)⁸⁹, while Chini et al. found a steepening (0–2 weeks)³⁸, Witteveen et al. showed no change (term infants 0–4 months)⁹⁰ and McSweeney et al. showed that aperiodic activity across childhood manifests as a quadratic change from 4 to 11 years old⁹¹. More longitudinal research in large populations of typically developing children is needed to clarify our understanding of aperiodic activity throughout development.

Clinical relevance of aperiodic activity

Finally, aperiodic activity may serve as a valuable biomarker for disease state. However, inconsistent results across studies makes interpretations challenging. A recent review explores the clinical relevance of aperiodic neural activity, highlighting the fact that aperiodic activity has been studied in the context of at least 35 distinct clinical diagnoses including Parkinson's disease, attention deficit hyperactivity disorder, autism, Alzheimer's disease, disorders of consciousness, depression and schizophrenia⁴. Of the diagnoses highlighted, studies linking aperiodic activity to Parkinson's disease, epilepsy and disorders of consciousness showed the greatest consistency across publications. Eighty per cent of the studies reporting a group-level comparison of the aperiodic exponent between individuals diagnosed with Parkinson's disease and control individuals reported an increased aperiodic exponent (steeper spectra). Studies on epilepsy consistently associate a greater aperiodic exponent with epileptic activity and foci. Studies on disorders of consciousness generally associate an increased aperiodic exponent with pathology. That being said, findings linking aperiodic activity to attention deficit hyperactivity disorder, autism, Alzheimer's disease, depression and schizophrenia did not show clear and consistent results across publications. The review highlights the subtle nuances that may contribute to these differences, but similar to the literature on the functional relevance of aperiodic activity, more work is needed to clarify the role of aperiodic activity.

Concluding remarks

Recent methodological developments have spurred interest in aperiodic neural activity, its physiological generators and its potential clinical and functional importance. Aperiodic activity—irregular, non-oscillatory fluctuations in neural field potentials—has many generators but primarily reflects asynchronous postsynaptic potentials across populations of neurons. Several methods have been developed to measure aperiodic activity, including spectral parameterization and IRASA, which both analyse the power spectra of EEG/LFP recordings. With these methods, researchers are beginning to shed light on the functional role and physiological basis of aperiodic activity. (See Box 2 for a review of open questions).

Electrophysiology research has traditionally emphasized neural oscillations and their role in cognition and disease. We highlight

BOX 2

Outstanding questions

• What are the generative mechanisms of aperiodic activity?

Computational modelling has provided insight into potential generative mechanisms of aperiodic activity. Empirical evidence supports the model that the aperiodic exponent reflects excitation–inhibition balance under certain conditions; however, many neural and non-neural processes contribute to neural field potentials. More work is needed to understand the relative contributions of these various processes, including calcium spikes, neuromodulatory currents, spiking activity, synchrony and non-neuronal sources of noise.

• What is the optimal method for quantifying aperiodic activity?

Several methods exist for measuring aperiodic activity, including irregular-resampling auto-spectral analysis (IRASA) and spectral parameterization, but each has advantages and limitations. There have been proposed methods to address limitations of spectral parameterization⁹⁵ and emerging methods that leverage components of both IRASA and spectral parameterization³⁰. One existing area of potential development in spectral parameterization is the automation of hyperparameter tuning to reduce manual intervention. Future research should investigate which of these methods has the most explanatory value with respect to physiology and behaviour.

• What is the appropriate timescale over which to quantify aperiodic activity?

Most studies to date have measured aperiodic activity during the resting state or longer task epochs; however, a few recent studies have begun investigating time-resolved estimates of aperiodic activity. This emerging research area shows that aperiodic activity is dynamically modulated during tasks, on the timescale of behaviour and cognition. Future research in this area may advance our understanding of the functional role of aperiodic activity.

• Do aperiodic dynamics have a functional role in neural communication and computation through something like a communication through aperiodic dynamics process?

Many influential theories of neural communication—for example, communication through coherence⁷⁹, binding by synchrony⁹⁶ and cross-frequency coupling⁹⁷—highlight the functional role of neural oscillations; however, aperiodic activity has not yet been integrated into these frameworks. Given that aperiodic dynamics overlap in time and space with those of oscillations, integrating theories of aperiodic activity into these frameworks may improve their explanatory value. In this Review, we introduce a theory of communication through aperiodic dynamics, which can be tested and expanded through targeted empirical studies.

recent empirical results that show event-related and stimulus-driven changes in aperiodic activity, independent of oscillations. Given the preponderance of evidence that aperiodic activity is dynamic and task-relevant, we argue that nearly all previous EEG and LFP studies have conflated neural oscillations with aperiodic activity, calling into question which results reflect true oscillations and which reflect aperiodic activity. This is because of the fallacious assumption that any power within a frequency band necessarily means that there is an oscillation at that frequency, known as the “Fourier fallacy”⁹². We do not argue that oscillations are not also functional and task-relevant. Rather, we strongly believe that we must revisit and re-evaluate classic

results whose claims regarding the function and physiology of putative oscillations do not account for aperiodic dynamics. This is because the results in the vast extant oscillations literature that does not account for aperiodic activity cannot, with any certainty, be ascribed solely to oscillatory processes.

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Author contributions

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Competing interests

The authors declare no competing interests.

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